

Bibliography (Selected Refereed Publications) – Zong-Liang Yang (As of 03/15/2025)

Google Scholar: <https://scholar.google.com/citations?user=ZncpGB0AAAAJ&hl=en>

Total Number of Papers Published = **230**; total citations = **35,881**; h-index = **81**

81 papers in AGU journals (Total number/Number authored by students or postdocs): JGR-Atmo (49/24), GRL (15/9), JAMES (6/5), WRR (4/4), Earth & Space Science (2/2), JGR-Earth Surface (1/1), JGR-Biogeosciences (2/0), JGR-Solid Earth (1/0), Earth's Future (1/0).

115 publications as senior/corresponding author (#), or with **graduates** (‡) & **postdoc advisees** (†).

Unraveling the Essential Mechanisms of Land Processes and Their Impact on the Earth System

1. **#Yang, Z.-L.**, et al., 1995: Preliminary study of spin-up processes in landsurface models with the first stage data of Project for Intercomparison of Land Surface Parameterization Schemes Phase 1(a), *J. Geophys. Res.*, **100 (D8)**, 16,553–16,578. (198 citations)
2. **#Yang, Z.-L.**, et al., 1997: Validation of the snow sub-model of the Biosphere-Atmosphere Transfer Scheme with Russian snow cover and meteorological observational data, *Journal of Climate*, **10**, 353–373. (384 citations)
3. **#Yang, Z.-L.**, et al., 1999: Sensitivity of ground heat flux to vegetation cover fraction and leaf area index, *J. Geophys. Res.*, **104 (D16)**, 19505–19514. (55 citations)
4. **#Yang, Z.-L.**, 2004: Modeling land surface processes in short-term weather and climate studies, in *Observations, Theory, and Modeling of Atmospheric Variability*, (ed. X. Zhu), World Scientific Series on Meteorology of East Asia, Vol. 3, Singapore, pp. 288–313. (70 citations)
5. ‡Cai, X., **#Z.-L. Yang**, J. B. Fisher, X. Zhang, M. Barlage and F. Chen, 2016: Integration of nitrogen dynamics into the Noah-MP land model v1.1 for climate and environmental predictions. *Geosci. Model Dev.*, **9**, 1–15, doi:10.5194/gmd-9-1-2016. (38 citations)
6. ‡David, C. H., D. R. Maidment, †G.Y. Niu, **Z.-L. Yang**, et al., 2011: River network routing on the NHDPlus dataset, *Journal of Hydrometeorology*, **12**, 913–934. (257 citations)
7. ‡Gulden, L.E., ‡E. Rosero, **#Z.-L. Yang**, et al., 2007: Improving land-surface model hydrology: Is an explicit aquifer model better than a deeper soil profile? *Geophys. Res. Lett.*, **34**, L09402, doi:10.1029/2007GL029804. [AGU Journal Editor's Highlight] (97 citations)
8. ‡Jiang, X.Y., C. Wiedinmyer, F. Chen, **#Z.-L. Yang**, and †J. C. F. Lo, 2008: Predicted impacts of climate and land-use change on surface ozone in the Houston, Texas, area, *J. Geophys. Res.*, **113**, D20312, doi:10.1029/2008JD009820. (131 citations)
9. ‡Knebl, M.R., **#Z.-L. Yang**, K. Hutchison, and D.R. Maidment, 2005: Regional scale flood modeling using NEXRAD rainfall, GIS, and HEC-HMS/RAS: A case study for the San Antonio River Basin summer 2002 storm event, *Journal of Environmental Management*, **75**, 325–336. (744 citations)
10. ‡Li, L. C., **#Z.-L. Yang**, et al., 2021: Representation of plant hydraulics in the Noah-MP land surface model: Model development and multiscale evaluation, *J. Adv. Model. Earth Syst.*, **13**, e2020MS002214. (101 citations; [Wiley top cited and top downloaded article in 2021–2022](#))
11. †Niu, G.-Y., **Z.-L. Yang**, R.E. Dickinson, and ‡L.E. Gulden, 2005: A simple TOPMODEL-based runoff parameterization (SIMTOP) for use in global climate models, *Journal of Geophysical Research*, **110**, D21106, doi:10.1029/2005JD006111. (503 citations)
12. †Niu, G.-Y. and **Z.-L. Yang**, 2006: Effects of frozen soil on snowmelt runoff and soil water storage at a continental scale, *Journal of Hydrometeorology*, **7 (5)**, 937–952. (528 citations)
13. ‡Parajuli, S. P., **#Z.-L. Yang**, and G. Kocurek, 2014: Mapping erodibility in dust source regions based on geomorphology, meteorology, and remote sensing, *J. Geophys. Res.–Earth Surface*, **119**, 1,977–1,994, DOI: 10.1002/2014JF003095. (95 citations)
14. ‡Parajuli, S. P., et al. (including **#Z.-L. Yang**), 2016: New insights into the wind–dust relationship in sandblasting and direct aerodynamic entrainment from wind tunnel experiments, *J. Geophys. Res.–Atmos.*, **121**, 1776–1792. (50 citations)

Developing Innovative Multi-physics-based Land Surface Models

15. #Yang, Z.-L. and †G.-Y. Niu, 2003: The Versatile Integrator of Surface and Atmosphere Processes (VISA) Part 1: Model description, *Glob. Planet. Change*, **38**, 175–189. (138 citations)
16. †Niu, G.-Y., #Z.-L. Yang, et al., 2011: The community Noah land surface model with multiparameterization options (Noah-MP): 1. Model description and evaluation with local-scale measurements, *J. Geophys. Res.*, **116**, D12109, doi:10.1029/2010JD015139. (2,591 citations)
17. #Yang, Z.-L., et al., 2011: The community Noah land surface model with multiparameterization options (Noah-MP): 2. Evaluation over global river basins, *J. Geophys. Res.*, **116**, D12110, doi:10.1029/2010JD015140. (741 citations)
18. Gupta, H.V., et al. (Z.-L. Yang), 1999: Parameter estimation of a land surface scheme using multicriteria methods, *J. Geophys. Res. Atmos.*, **104 (D16)**, 19491–19503. (368 citations; Dr. Yang spearheaded the pioneering application of multicriteria methods to land surface modeling.)
19. Henderson-Sellers, A., Z.-L. Yang and R.E. Dickinson, 1993: The Project for Intercomparison of Land-surface Parameterization Schemes, *Bull. Am. Meteorol. Soc.*, **74**, 1335–1349. (574 citations; Dr. Yang's dissertation article sparked the 20-year PILPS project internationally.)
20. ‡Jiang, X., †G.-Y. Niu, and #Z.-L. Yang, 2009: Impacts of vegetation and groundwater dynamics on warm season precipitation over the Central United States, *J. Geophys. Res.*, **114**, D06109, doi:10.1029/2008JD010756. (151 citations)
21. ‡Rosero, E., #Z.-L. Yang, et al., 2010: Quantifying parameter sensitivity, interaction and transferability in hydrologically enhanced versions of Noah-LSM over transition zones, *J. Geophys. Res.*, **115**, D03106, doi:10.1029/2009JD012035. (183 citations)

Integrating Land Modeling, Remote Sensing, Climate Modeling, and Water Resource Management

22. #Yang, Z.-L., R.E Dickinson, 1996: Description of the Biosphere-Atmosphere Transfer Scheme for Soil Moisture Workshop and performance evaluation, *Glob. Planet. Change*, **13**, 117–134 (134 citations)
23. †Xu, Z.F., #Z.-L. Yang, 2012: An improved dynamical downscaling method with GCM bias corrections and its validation with 30 years of climate simulations, *J. Clim.*, **25**, 6271–6286. (193 citations)
24. †Chen, W.-J., C.-L. Huang, Z.-L. Yang, 2021: More severe drought detected by the assimilation of brightness temperature and terrestrial water storage anomalies in Texas during 2010–2013, *Journal of Hydrology*, **603**, 126802, <https://doi.org/10.1016/j.jhydrol.2021.126802> (11 citations)
25. Gent, P. R., et al. (including Z.-L. Yang), 2011: The Community Climate System Model Version 4, *J. Climate*, **24**, 4973–4991, doi: 10.1175/2011JCLI4083.1. (3,176 citations; Dr. Yang is a prominent innovator and contributor in the development of the land component of the climate model.)
26. ‡Li, L. C., D. X. She, H. Zheng, P. R. Lin, and #Z.-L. Yang, 2020: Elucidating diverse drought characteristics from two meteorological drought indices (SPI and SPEI) in China, *Journal of Hydrometeorology*, **21 (7)**, 1513–1530. (211 citations)
27. Nielsen-Gammon, J. W., et al., 2020: Unprecedented drought challenges for Texas water resources in a changing climate: what do researchers and stakeholders need to know? *Earth's Future*, **8**, e2020EF001552. (105 citations; Dr. Yang spearheaded numerous Water Forums and Discussion Groups, engaging water resource managers to tackle escalating droughts.)
28. †Niu, G.-Y., Z.-L. Yang, et al., 2007: Development of a simple groundwater model for use in climate models and evaluation with Gravity Recovery and Climate Experiment data, *J. Geophys. Res.*, **112**, D07103, doi:10.1029/2006JD007522. (652 citations)
29. ‡Wu, W.-Y., ..., and Z.-L. Yang, 2020: Divergent effects of climate change on future groundwater availability in key mid-latitude aquifers, *Nature Communications*, **11 (1)**, 3710, DOI: 10.1038/s41467-020-17581-y. (348 citations)
30. ‡Zhao, L. and #Z.-L. Yang, 2018: Multi-sensor land data assimilation: Toward a robust global soil moisture and snow estimation, *Remote Sensing of Environment*, **216**, 13–27. (52 citations)